

No. of Functional Features Included in the Solution

Date	7 July 2026
Team ID	SWTID-2026-8075
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Speech-to-Text Conversion	Converts uploaded audio into text using AI.	Python (Whisper Module)	Done	High
2	Concept Selection	Allows users to select the concept before analysis.	Python (Streamlet UI)	Done	Medium
3	Semantic Similarity Analysis	Compares transcribed text with the reference answer and calculates similarity score.	Python (similarity.py)	Done	High
4	Fluency Analysis	Calculates speaking duration, word count and Words Per Minute (WPM).	Python (Libros)	Done	High
5	Pause Detection	Detects silence in speech and calculates pause ratio.	Python (NumPy + Libros)	Done	Medium
6	Voice Energy Analysis	Measures RMS energy to evaluate voice confidence.	Python (Libros)	Done	Medium
7	Filler Word Detection	Detects filler words such as <i>um, uh, actually</i> , etc.	Python	Done	Medium
8	PDF Report Generation	Generates a detailed analysis report in PDF format.	Python (pdf_report.py)	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	10
Total Features Implemented	10
Core / Must-Have Features	Speech-to-Text, Semantic Similarity, Fluency Analysis, PDF Report Generation

Additional / Nice-to-Have Features	Pause Detection, RMS Energy Analysis, Filler Word Detection, Audio Waveform Visualization, AI Performance Summary, Understanding Level Classification
Features Tested & Verified	Speech-to-Text, Similarity Score, Fluency Analysis, Pause Detection, RMS Energy Analysis, Filler Word Detection, Waveform Display, PDF Report Generation

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	4	Concept Selection, Audio Upload, Analysed Button, Download PDF
2	Backend / Logic	6	Speech-to-Text, Semantic Similarity, Fluency Analysis, Pause Detection, RMS Energy Analysis, AI Feedback
3	Database / Storage	1	Audio File Storage (uploads folder)
4	API / Integration	1	OpenAI Whisper Integration
5	Security / Authentication	0	Not Implemented